ABC MANUFACTURING LTD.

Factory Operations & Equipment Register | Document ID: OPS-ABC-001 | Revision 1.0 | Effective: 01 Nov 2026

Document purpose: This document describes the fictional factory layout, equipment, production operations, operating constraints, dependencies and maintenance windows. It is intended as an OCR-ingestion test document for GridVision's Commercial Operations Agent.

1. Factory profile

Field	Value
Site	ABC Manufacturing Ltd. — Plant 1
Location	Pune, Maharashtra, India (demonstration location)
Operating pattern	Three shifts: A 06:00–14:00; B 14:00–22:00; C 22:00–06:00
Production days	Monday–Saturday; Sunday reserved for planned maintenance where possible
Electricity supply	Grid connection; rooftop solar is not yet installed
Production target	1,000 finished units per operating day (planning assumption)
Document owner	Plant Operations Manager

2. Production areas and machine register

The rated power values below are nameplate values. Actual consumption varies with load, duty cycle, idle time and operating condition.

Area	Machine ID	Equipment	Rated power	Normal operating window	Notes
Production	CUT-01	CNC cutting machine	45 kW	06:00–22:00	Feeds machined components to CNC-02
Production	CNC-02	CNC machining centre	60 kW	06:00–22:00	Requires CUT-01 output; setup changeover: 30 min
Production	ASM-01	Assembly line	22 kW	06:00–22:00	Requires machined components from CNC-02
Heating	FUR-01	Heat-treatment furnace	120 kW	00:00–24:00	Batch process; preheat approximately 90 min
Cooling	CHL-01	Central chiller	75 kW	00:00–24:00	Cooling demand rises during high ambient temperature
Material handling	CMP-01	Air compressor	37 kW	00:00–24:00	Supplies compressed air to production
Material handling	CON-01	Main conveyor	15 kW	06:00–22:00	Runs when production line is active
Utilities	PMP-01	Process water pump	18.5 kW	00:00–24:00	Minimum flow must be maintained
Utilities	LGT-01	Plant lighting	12 kW	As required	Non-production zones can be switched off by zone

3. Process sequence and dependencies

Sequence	Operation	Input / prerequisite	Output / next dependency
1	CUT-01 cutting	Raw material available; operator assigned	Cut blanks → CNC-02
2	CNC-02 machining	Cut blanks from CUT-01; tooling available	Machined components → ASM-01
3	FUR-01 heat treatment	Approved batch loaded; recipe selected	Heat-treated components → inspection / assembly
4	ASM-01 assembly	Machined and heat-treated components available	Finished units
5	CON-01 material movement	Production line active	Transfers material between designated work areas

Operational dependency rules

• CNC-02 must not start a production batch until the required CUT-01 blanks and approved tooling are available.
• ASM-01 must not be scheduled ahead of the availability of its required machined components.
• FUR-01 batch recipes and temperature limits are controlled by the approved process sheet; operators must not change them to reduce energy cost.
• CHL-01 and PMP-01 must maintain process conditions. They are not to be switched off solely to reduce electricity consumption.
• CMP-01 pressure must remain within the approved operating band. Leak repairs and pressure optimization require maintenance authorization.

4. Operating constraints and flexibility

Asset / process	Constraint	Permitted flexibility
CUT-01 / CNC-02	Daily production target and downstream sequence must be met	Start time may move by up to 2 hours with supervisor approval
ASM-01	Must receive components before assembly	Can be shifted within A/B shifts; no unsupervised night shift
FUR-01	Batch recipe, ramp rate and soak time are quality-critical	Batch start can move by up to 3 hours if delivery plan permits
CHL-01	Maintain required chilled-water temperature and flow	Setpoint changes only within approved range and with facilities approva
CMP-01	Maintain minimum line pressure	Leak repair / pressure changes require maintenance approval
PMP-01	Maintain minimum process-water flow	No planned shutdown during production

5. Maintenance windows and restrictions

Machine ID	Maintenance activity	Preferred window	Restriction
FUR-01	Burner inspection and refractory check	Sunday 08:00–14:00	Do not interrupt an active batch
CHL-01	Condenser and filter inspection	Sunday 09:00–12:00	Facilities team approval required
CMP-01	Leak survey and filter service	Sunday 10:00–13:00	Maintain backup air supply before isolation
CNC-02	Spindle and lubrication inspection	Sunday 08:00–11:00	Lockout/tagout procedure mandatory
PMP-01	Seal and coupling inspection	Sunday 11:00–13:00	Confirm alternate water supply before isolation

6. Illustrative operating observations

Observation ID	Observation	Potential energy relevance	Status
OBS-01	CHL-01 appears to operate at high load during afternoon cooling-demand peaks	Review cooling-demand response and condenser cleaning	To be investigated
OBS-02	CMP-01 has reported air leakage near the packaging area	Compressed-air losses may increase compressor runtime	Maintenance ticket required
OBS-03	LGT-01 lighting remains on in two storage zones after Shift C	Potential avoidable off-shift load	Verify occupancy and safety requi
OBS-04	FUR-01 batch start time varies between production days	Batch scheduling may affect exposure to tariff periods	Coordinate with production planni

7. Data and authority notes

All machine ratings, process details, operating windows, observations and constraints in this document are fictional demonstration values created for GridVision development. They are not measured telemetry, approved engineering limits or instructions for real equipment. The Operations Agent must treat them as document-derived context, retain document/page provenance, and request confirmation before recommending changes that affect safety, product quality or production.

Suggested OCR extraction targets: factory/site name; machine IDs; equipment names; rated power; operating windows; process dependencies; operating constraints; maintenance windows; observation IDs; and source page references.